(a) All seven $\pi \rightarrow \pi^*$ states
C001 Furofuran
C002 Furofuran
I002 Criegee
I004 Benzoxadiazole
I005 Hexatriene
V010 Furan
V011 uracil
−0.5
0.0
0.5
Signed excitation error / eV
MAE 0.180 eV; max 0.349 eV
(b) Thirteen $n \rightarrow \pi^*$ controls
I001 Criegee
N001 acetaldehyde
N002 formamide
N003 pyrazine
V001 acetone
V002 glyoxal
V003 tetrazine
V004 acrolein
V005 nitrosomethane
V006 maleimide
V007 pyrimidine
V008 cyclopentadienone
V009 pyranone
−0.4
−0.2
0.0
0.2
0.4
Signed excitation error / eV
MAE 0.250 eV; max 0.498 eV
Previous compact space
Screened symmetry augmentation
Fixed 50/50 reference weights; lambda = 0.9; no energy offset. All twenty states are exposed development.